

ELEC497: Multi-Channel EEG Recording With NESS

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Abstract — Traditional electroencephalography systems, even modern wireless variants, remain visually obtrusive and reliant on centralized architectures. This report details the advancement of the Neuro-electrical Scalp Sensor, an imperceptible, decentralized wireless sensor network. The study focuses on optimizing the fabrication of the Carbon Nanotube composite electrodes used within this sensor, validating multi-channel functionality, and implementing a firmware-based time synchronization protocol. Fabrication refinements, including a 2.51 weight percent concentration and an integrated copper molding technique, successfully reduced base resistance of electrodes below 40 kilo-ohms and streamlined the fabrication process, while a 10-minute, 5-volt silver-plating process further reduced resistance to under 5 kilo-ohms. While functional testing successfully captured visual P300 event-related potentials, it demonstrated significant temporal drift between the independent sensors. To address this challenge, a two-stage synchronization framework was developed. Using localized communication exchanges and sliding-window linear regression, the protocol dynamically aligns independent node clocks to a central receiver and subsequently to the host device's local clock. This work resolves long term drift misalignment and provides future directions for further synchronization improvements.

Keywords—*Visually Imperceptible Electroencephalography Sensor, Bluetooth Low Energy Node Synchronization*

I. INTRODUCTION

Electroencephalography (EEG) is a widely used method for monitoring the brain's electrical activity. It is done by attaching electrodes on the scalp, and recording the microvolt fluctuations generated by the summed postsynaptic potentials of firing brain cells [1]. EEG provides a non-invasive window into cognitive processes, mental health, and human-machine interaction [2], [3]. Despite these benefits, traditional EEG acquisition systems are notoriously bulky and difficult to deploy in real-world environments. While wireless headsets have introduced some mobility, they remain visually conspicuous and ergonomically limited, hindering their application for monitoring in everyday environments [4].

While significant strides have been made to reduce the size of individual sensors, the industry lacks socially discreet EEG systems that maintain high signal fidelity without the need for a bulky processing hub [5], [6]. Furthermore, the fabrication of flexible, skin-conforming electrodes that can maintain stable impedance with minimal conductive gel application remains a challenge [6]. This issue was tackled within Dr. Ameri's nano research group, where a Neuro-electrical Scalp Sensor (NESS) was constructed using Carbon Nanotubes (CNT), Polydimethylsiloxane (PDMS), and Silicone Oil (SO), and

compared to gold standard EEG recording devices with near identical signal integrity [7]. However, multi-channel EEG recording has not yet been demonstrated with these devices.

Building on this foundation, the current project aims to replicate the fabrication process to construct additional NESS devices and attempt multi-channel recording. The research was conducted in three distinct stages. First, the sensor fabrication process was optimized, focusing on refining parameters such as CNT concentration levels and silver-plating voltage and duration to improve electrical conductivity of the electrodes and reduce surface flaking. Second, the functional performance of the assembled sensors was validated through multi-channel EEG recordings on human subjects. Finally, the project addressed the technical challenge of multi-nodal synchronization by implementing a microprocessor-driven communication protocol, aligning the local clocks of all existing nodes to a central hub.

II. RELATED WORK

Recent years have seen significant strides in the development of wireless recording devices. A comprehensive survey by Niso et al. compared 48 different wireless EEG systems, including current state-of-the-art platforms capable of supporting up to 32 channels, such as g.tec's g.Nautilus PRO Flexible, and Mentalab's Explore Pro [8]. This comparison demonstrated that despite the shift toward wireless data transmission, most of these systems remain visually obtrusive and ergonomically bulky. The primary limitation stems from their reliance on a centralized architecture, where individual electrodes must still be physically tethered to a head-mounted amplifier [8]. While this centralized design successfully eliminates temporal misalignment by utilizing a shared sampling clock, the routing of cables across the scalp restricts wearability and leaves the system vulnerable to motion artifacts caused by cable movement [9]. Consequently, these established systems fall short of the requirements for seamless, naturalistic monitoring.

Comparatively, NESS falls into a more recent category of wireless EEG sensors known as Wireless EEG Sensor Networks (WESN). A notable example of this architecture is the EEG-Linx platform [9], which shares NESS's goal of enabling mobile, unobtrusive monitoring during daily activities. Both systems achieve this through a decentralized, modular design that replaces bulky centralized hubs with independent, free-standing sensor-circuit nodes. While both platforms successfully use this new approach, NESS pushes the boundaries of visual imperceptibility. Linx nodes are described as unobtrusive, measuring approximately 2 by 3 cm and supporting up to four channels per

node, whereas NESS devices are made up of two 8.5 mm diameter circular nodes, spaced approximately 5 cm apart, enabling them to hide under hair strands. Furthermore, NESS utilizes a specialized silver-plated CNT/PDMS/SO composite electrode with a micro-suction cup array, providing superior wearability and signal fidelity. However, without a shared hardware clock, wireless communications received from WESN systems are prone to temporal misalignment. Addressing the resulting clock drift and packet jitter therefore requires robust software-based synchronization protocols. The EEG-Linx platform addresses this by employing a custom software protocol that dynamically estimates wireless link latency and adjusts timestamps relative to a reference node to compensate for oscillator drift [9]. While this ensures sample-level synchronization for Linx devices, the current iteration of the NESS architecture lacks a comparable time-alignment mechanism. Consequently, implementing a robust drift synchronization protocol for NESS forms a central objective of this report.

The challenge of synchronizing decentralized wireless nodes is not unique to EEG systems; rather, it represents a foundational problem in the broader field of Wireless Sensor Networks (WSNs). Influential works in this domain have already been exploring software-based solutions to achieve high-precision time alignment across independent oscillators. For instance, in 2002, Elson et al. introduced Reference-Broadcast Synchronization (RBS), a classic framework that achieves microsecond-level precision by utilizing physical-layer broadcasts to eliminate sender-side nondeterminism [10]. Similarly, the Flooding Time Synchronization Protocol (FTSP), developed by Maróti et al. in 2004, is a widely cited benchmark protocol for robust synchronization using medium access control (MAC) layer time-stamping and continuous clock skew estimation [11].

While our implemented ping-pong framework shares core mathematical principles with these foundational models, such as FTSP's use of linear regression to estimate clock skew, it diverges significantly in its structural execution. Unlike RBS, which relies on receiver-receiver synchronization via reference broadcasts, our synchronization protocol employs a strict master-slave topology between the ESP32 and the nrf51822's on each NESS device. And whereas FTSP attains microsecond-level precision through low-level MAC layer timestamping in a root-based flooding architecture, our system operates at the application layer using Bluetooth Generic Attribute Profile (GATT) characteristics. To address the transmission jitter introduced by higher-level Bluetooth communication, our framework uses a 15-point sliding-window regression to take the global average time delay over a 150 second timeframe.

III. OVERVIEW OF APPROACH

The initial phase of the project focused on the construction of additional NESS devices. Each device consists of two electrodes (one for acquiring the signal and the other used for providing a baseline reference), an integrated battery, and two printed circuit boards (PCBs) for Bluetooth communication and signal amplification. The standard manufacturing procedure for the electrodes consists of five

stages: formulating the precursor mixture, casting the sugar molds, preparing the precursor dipping basin, performing the dipping procedure, and silver plating the electrode's surface. Starting with the precursor mixture, for a total composite of 10 g, 0.251 g of CNT (2.51 wt%), 7.04 g of PDMS (70.4 wt%), 2 g of SO (20 wt%), and 125 mL of isopropyl alcohol (IPA) were mixed in a beaker (with the PDMS curing agent accounting for the final 7.09 wt%). This solution was subjected to multiple rounds of manual and ultrasonic mixing; throughout this process, CNT particles were actively scraped from the beaker walls to prevent material loss, and any visible agglomerates were mechanically dispersed using a spatula. The solution was then heated on a hotplate for three days until IPA fully evaporated. With the IPA extracted, the precursor was then ready for the addition of the curing agent, which is explained in a later step.

With the precursor mixture completed, the sugar molds were then created. This process required strict thermal regulation. The sugar was heated in a covered Petri dish up to 210°C until it achieved a golden-orange hue, while the soft mold used to shape the sugar was preheated in a separate dish to 190°C. Upon the onset of bubbling in the sugar, the cover was removed, and the hotplate temperature was reduced to 200°C. The liquid sugar was then cast onto the soft mold and pushed into the holes using a thin scalpel to remove any leftover bubbles.

The subsequent stage involved preparing a uniform layer of the precursor mixture. A customized basin was constructed on a glass slide using a double layer of adhesive tape to create a cavity approximately 15 mm in length, 15 mm in width, and 4 mm in depth (each layer of tape being 2 mm thick). This allowed a copper strip to be embedded directly in between the layered tape. 1 g of precursor mixture was placed in a small mixing cup. Then 0.2 g of PDMS curing agent was mixed into the precursor and the result was dispensed and leveled within the basin, surrounding the copper strip. To remove entrapped air, the mixture was debubbled for two 20-minute intervals, with the surface being re-leveled after each cycle.

The physical molding process concluded by attaching the sugar mold to a glass slide with epoxy. To ensure proper planar alignment, a secondary glass slide was temporarily balanced atop the sugar mold to provide clear visual feedback, and leveling adjustments were executed using a blunt-tipped tweezer. Once the epoxy hardened, the sugar mold was mounted on the x, y, z linear stage, decreasing the linear stage in the z direction until one third of the diameter of the sugar beads was fully immersed into the leveled precursor. The stage was subsequently increased in the z direction to form the suction cup structures. The configuration was then heated at 80°C for 24 hours for curing. Finally, the cured electrodes were silver-plated at 5 V for 10 minutes and submerged in water for 5 hours to dissolve the residual salt buildup from the plating process.

During final assembly, two completed electrodes were selected, and their copper strips were connected to the PCBs using soldered wires. One electrode was secured to the node housing the lithium-ion battery, and the other to the node containing the communication and amplification PCBs, rendering the device ready for testing.

The final phase of the project focused on the implementation of a long-term drift correction framework to precisely synchronize multi-channel data across the wireless sensor network. In the established architecture, individual sensor nodes utilize an nRF51822 System-on-Chip to transmit recorded data via Bluetooth Low Energy (BLE) to a Saeed Studio XIAO ESP32 central receiver. Because standard packet timestamping upon arrival fails to account for variable Bluetooth latency and intrinsic hardware clock drift, a continuous, two-stage synchronization protocol was developed.

The first stage corrects desynchronization between the nodes and the ESP32 using a localized "ping-pong" communication exchange. The ESP32 initiates this sequence by dispatching a separate "ping" packet to each node at a ten second interval. Once the nRF51822 receives the request, it immediately captures its 32-bit local microsecond timestamp (T_{node}) and returns it as a "pong" packet. The ESP32 records the exact departure (T_1) and arrival times (T_2) of this exchange, transmitting all three recorded times to the Python script on the host computer. Assuming symmetrical Bluetooth latency, the Python framework calculates the master time corresponding to the node's timestamp by adding half the round-trip time to T_1 . The instantaneous offset is then derived by subtracting T_{node} from this estimated master time. To convert these singular data points into a robust synchronization model, the framework tracks their temporal differential over time using linear regression ($y = mx + b$). This maps each node's independent clock relative to the ESP32's master timeline, dynamically computing the ongoing drift rate (m) alongside the initial startup offset (b). To average out random Bluetooth transmission jitter while remaining adaptable to non-linear oscillator drift, this regression calculation is strictly limited to a sliding window of the 15 most recent synchronization points.

The second stage finds the difference between the ESP32's master time and the local clock of the computer. Upon receiving a synchronization packet from the ESP32, the Python script immediately captures the current timestamp from the high-precision Lab Streaming Layer (LSL) clock. The instantaneous transmission offset is calculated by subtracting the ESP32's timestamp from this LSL timestamp. To filter out operating system-level processing jitter, these raw offsets are continuously averaged across a 20-point sliding window. By adding this stabilized offset to the incoming data streams, the framework translates the EEG samples from the microcontroller's timeline to the computer's timeline. This final alignment ensures that every sample across all discrete wireless channels is accurately unified on a single temporal continuum.

IV. EXPERIMENT DESIGN

Throughout the electrode fabrication process, electrical resistance was verified using a standard digital multimeter. This validation occurred at two critical stages: first, during the initial precursor optimization to evaluate the resistance of the cured CNT/PDMS/SO composite; and second, to quantify the reduction in surface resistance achieved by the silver-plating process. To ensure measurement accuracy, readings were recorded only after the instrument had stabilized to a steady-state value, accounting for the inherent settling time of the electrode's surface resistance.

To evaluate the durability of the sensors and silver coating over repeated use, an Electrode-Skin Interface Impedance (ESII) test was conducted using an LCR meter. A wire was soldered to the embedded copper strips of two electrodes. One electrode was connected via alligator clip to the positive terminal of the LCR meter, and the other to the ground terminal. A minimal layer of conductive gel was applied to the suction-cup interface of both electrodes, which were then placed approximately 2 cm apart on the subject's forearm. The LCR meter performed a frequency sweep from 20 Hz to 1000 Hz. To simulate the degradation of continuous wear and cleaning cycles, the electrodes were removed, thoroughly cleaned, and allowed to dry completely before repeating the measurement. This exact procedure was executed consecutively for 10 trials to track impedance stability over time.

A multi-channel analysis was conducted using the constructed EEG devices, with their signals simultaneously compared against a gold-standard 32-channel EEG system from Brain Product's [12]. Two NESS devices were positioned on the subject's scalp at the Fz and POz locations, and the 32-electrode cap was placed over them. To elicit visual P300 responses, the subject watched a screen displaying an alternating checkerboard pattern at 2-second intervals. Event triggers were embedded in the recorded EEG data at each screen alternation to precisely mark the onset of the stimuli and reference the corresponding P300 pulses.

To quantitatively measure the effectiveness of the synchronization protocol, the long-term drift between two separate NESS devices were measured before and after the protocol was instilled. A standard function generator was used to generate a continuous 10 Hz, 5 mV peak-to-peak sine wave, which was simultaneously fed into both sensors for a duration of three hours and a half. The resulting data logs were then processed to quantify the drift offset between the two devices over time. To isolate the 10 Hz target signal from ambient noise, a 9–11 Hz bandpass filter was applied. The relative temporal drift was then calculated using IQ Demodulation. A Hilbert transform was applied to the filtered data of the first device to generate an analytic signal, allowing for the extraction of the real (In-phase, I) and imaginary (Quadrature, Q) components. The phase relationship between the two sensors was then obtained by multiplying these components by the signal from the second device. A 0.5 Hz low-pass filter was then applied to the resulting I and Q components to remove the high-frequency artifacts generated during the signal multiplication. The absolute clock drift in milliseconds was then mathematically derived using (1):

$$Drift (ms) = \frac{\arctan(\frac{Q}{I})}{2\pi} * 100 \quad (1)$$

This phase difference was plotted against the local recording time to visualize the drift throughout the entire recording.

V. RESULTS

During the initial fabrication phase, several optimizations were required to achieve the necessary mechanical and electrical properties for the NESS electrodes. Initial precursor batches utilizing a baseline 2.4 wt% CNT concentration yielded a cured

resistance between 500 k Ω and 10 M Ω , well above the desired resistance under 40 k Ω required for an optimal signal-to-noise ratio (SNR). Additionally, it was observed that significant material was being lost to the beaker walls and mixing implements between manual and ultrasonic mixing cycles. By increasing the baseline concentration to 2.5 wt% and introducing an additional 0.01 wt% of CNT to compensate for mixing losses (bringing the final concentration to 2.51 wt%), the optimized batch consistently achieved under 40 k Ω resistance while maintaining the flexible form necessary for suction cup adhesion.

Thermal regulation during sugar mold casting also required refinement. Excessive hotplate temperatures caused severe bubbling within the liquid sugar; these bubbles were difficult to extract and resulted in surface imperfections on the final mold. Establishing a 210°C/200°C stepped heating configuration to melt the sugar, while maintaining the soft mold at 190°C, provided optimal sugar consistency, ensuring sufficient viscosity for structural conformity while minimizing bubble formation.

Modifications to the precursor basin significantly streamlined the fabrication process of the electrodes. Embedding the copper strip directly into the uncured mixture by adding a second layer of tape to the basin eliminated the need for a secondary precursor application step, which had previously resulted in surface imperfections and added manufacturing overhead. The only minor complication introduced by this integrated method was the risk of damaging the copper strip when cutting the electrode out into a circular shape.

Finally, empirical testing of the silver-plating process determined that plating at 5 V for 10 minutes yielded optimal results, achieving a measured surface resistivity of approximately 5 k Ω and demonstrating robust coating durability. This durability is evidenced by ESII tests conducted over successive rounds of electrode cleaning. As illustrated in Figure 1: Electrode Skin Interface Impedance sweep from 20 Hz to 1000 Hz over 10 continuous cleaning cycles., the impedance remains relatively constant through the first seven cleaning cycles before increasing linearly. This long-term stability occurs because the optimized voltage and coating duration successfully mitigate the surface flakiness typically associated with the thicker silver films produced at higher voltages or longer plating times.

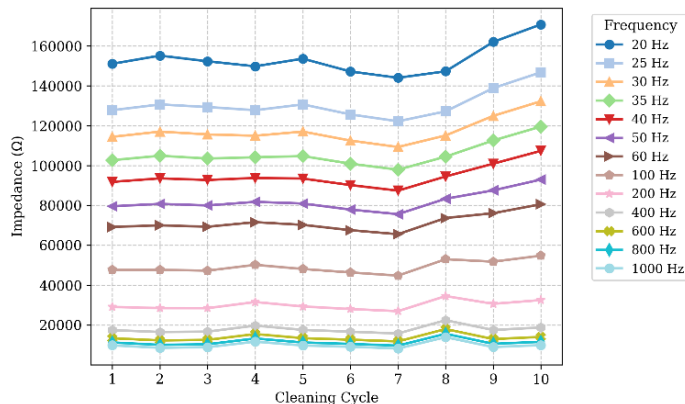


Figure 1: Electrode Skin Interface Impedance sweep from 20 Hz to 1000 Hz over 10 continuous cleaning cycles.

During the second phase of the project, the assembled sensors were deployed for multi-channel EEG data acquisition to capture visually evoked P300 event-related potentials. Although the hardware successfully acquired the biosignals, significant temporal drift between the individual electrode nodes became apparent over extended recording periods. This drift impacted the results of the P300 waveform. As seen in Figure 2, the signal recorded by the NESS device has a smaller P300 spike compared to the gold-standard system. Because these final signal representations were derived by averaging multiple P300 trials gathered over the entire session, slight temporal shifts in stimulus onset caused by the drift resulted in a suppressed average signal. This desynchronization was primarily attributed to minor variations in the local oscillator frequencies caused by non-idealities within the individual quartz crystals and exacerbated by localized temperature fluctuations.

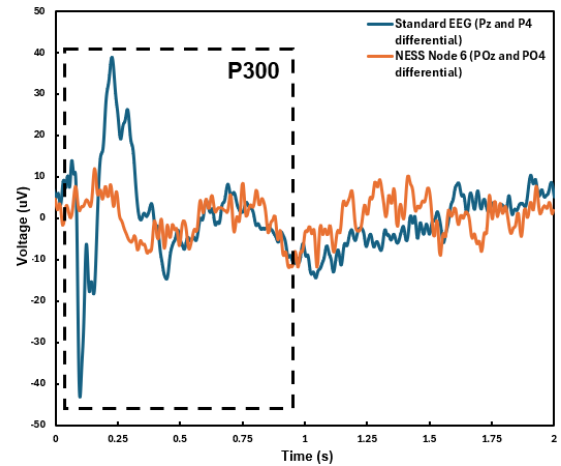


Figure 2: Plot comparing P300 signals of NESS and Gold Standard.

To properly visualize the temporal error introduced by these non-idealities, the drift between two NESS devices was monitored over a 3.5-hour period. Figure 3 and Figure 4 show the drift before and after the addition of the “ping-pong” synchronization protocol, respectively. As seen in Figure 3, the long-term drift between the two devices continuously oscillates up and down between -10 ms and 10 ms, increasing all the way to -15 ms by the end of the recording. The constant waveform in Figure 4 demonstrates that the protocol improves the stability of the long-term drift over time. However, it still struggles with short term jitter, as demonstrated by the spikes throughout the recording, reaching up to 15 ms difference.

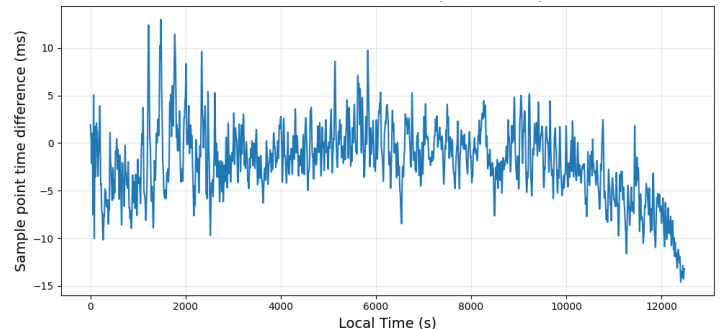


Figure 3: Drift between two NESS devices with no synchronization.

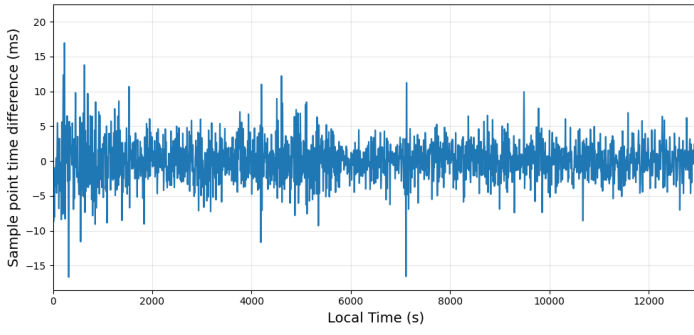


Figure 4: Drift between two NESS devices with synchronization.

VI. THREATS TO VALIDITY

Several threats to validity must be considered when interpreting the ESII and silver-plating durability results. First, the measured impedance could have been influenced by minor variations in exact electrode placement, inconsistencies in applied conductive gel volume, mechanical stress on the soldered interconnects, and progressive changes to the subject's skin hydration across the ten consecutive trials. Second, reliance on a single subject limits the generalizability of the findings, as anatomical differences in skin characteristics across broader populations may yield significantly different results. Finally, while the repeated cleaning cycles effectively evaluated the coating's resistance to mechanical abrasion, this test does not fully capture the complex degradation mechanics of continuous real-world wear, such as prolonged exposure to sweat, fluctuating temperatures, and varying environmental humidity.

While the long-term drift between the NESS network is now actively controlled, there remain several limitations with the current protocol. The primary limitation is the assumption of symmetric round-trip delay. The implemented “ping-pong” algorithm estimates instantaneous clock offset by dividing the measured round-trip time by two, thereby assuming identical forward and return delays. In BLE, however, packet transmission is governed by discrete connection intervals, and application-layer packets may be buffered until the next connection event; as a result, one-way transmission delays can vary substantially depending on connection-event timing and blocked transmissions [13]. To reduce the effect of this timing variability, the protocol applies a 15-point sliding-window regression over recent synchronization points. Although this reduces instantaneous variance, it does not fully remove offset errors when the forward and return path delays are unequal over several consecutive intervals.

Additionally, this demonstrates that the experimental process should also be expanded to validate the short-term jitter correction capabilities of the synchronization protocol. While plotting the long-term drift effectively highlights the systems capabilities at handling temperature fluctuations and local oscillator non-idealities, it does not provide a precise view into the short-term errors caused by the unpredictable Bluetooth queues. To ensure a more robust validation in future work, the experimental design should include an analysis of the point-to-point phase variance to quantify instantaneous error data.

Furthermore, tracking exactly when packets drop, and using a simultaneous physical electrical pulse to trigger both devices would help verify their exact timing alignment.

VII. CONCLUSION

This research significantly advances the development of the NESS device as a viable platform for imperceptible, decentralized wireless EEG monitoring. The study successfully optimized the fabrication of flexible CNT/PDMS/SO composite electrodes by establishing a 2.51 wt% CNT concentration, integrating a single-step copper molding technique, and applying a 5 V, 10-minute silver-plating process. These refinements enhanced mechanical durability and consistently reduced surface resistance to under 5 k Ω . While functional testing validated the hardware's capacity to capture P300 event-related potentials, it also verified that decentralized node architectures suffer from critical temporal drift.

To address this challenge, a two-stage synchronization protocol was implemented within the device firmware architecture. By utilizing a localized “ping-pong” communication exchange paired with sliding-window linear regression, the framework successfully bounds macroscopic long-term drift, aligning the independent nRF51822 oscillators to an ESP32 central receiver and the host computers local oscillator. However, the current iteration remains limited by the non-deterministic, asymmetrical transmission latency inherent to BLE connection queues, which introduces short-term jitter.

To resolve this limitation, future work must transition away from out-of-band ping-pong synchronization. Instead, the local nRF51822 microsecond hardware timestamp should be embedded directly into the 20-byte payload of every outgoing EEG data packet. By attaching the hardware timestamp directly to the physical data, the framework will completely bypass BLE transmission queue delays and eliminate the reliance on assumed symmetrical round-trip times. Coupled with an expanded experimental methodology, including analyzing unfiltered point-to-point phase variance to quantify short-term jitter and the use of simultaneous physical electrical pulses to verify absolute temporal alignment, these improvements will improve NESS's capabilities for accurate multi-channel EEG recording in naturalistic environments.

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